

FITPULSE 3D: REAL-TIME GYM MANAGEMENT & BIOMETRIC TELEMETRY SYSTEM

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Abstract

FitPulse is an enterprise-grade fitness ecosystem integrating physical biometric access control with automated financial and coaching governance. Built with React 19, Three.js, Node.js, and MySQL, the platform features 128-D neural facial recognition turnstiles, 60-second rotating TOTP MFA, and real-time WebSocket telemetry for athlete check-in alerts.

Main Outcome: Eliminates buddy-punching, unifies coach-athlete telemetry, and provides automated day-level prorated billing for mid-cycle plan switches with zero-refund policy enforcement.

Problem Statement & Objectives

Problem: Conventional fitness centers rely on shared RFID tags or manual desk registers vulnerable to unauthorized tailgate access, with frequent arithmetic errors in mid-cycle membership tier adjustments and coach allocations.

- Obj 1 (Biometric Access):** Implement dual-factor turnstile Face ID (128-D neural embeddings) paired with dynamic 60s rotating TOTP MFA passcodes.
 - Obj 2 (Prorated Billing):** Build an ACID-compliant billing engine with Cashfree PG integration, unused-credit upgrades, and strict non-refundable downgrade rules.
 - Obj 3 (Real-time Telemetry):** Establish full-duplex WebSockets for instant check-in notifications to assigned trainers and unified administrative governance.
- **Scope:** Role-based portals (Member, Coach, Admin), automated Cashfree checkout, workout programmer, nutrition logging, and durable audit logs.

Existing vs Proposed System

Feature	Existing Gym Systems	Proposed System (FitPulse 3D)
Access Control & Attendance	Static RFID / manual logs; prone to buddy punching & loss.	Dual-factor: 128-D Face-API.js + 60s rotating TOTP MFA.
Plan Upgrades	Manual front-desk estimations; frequent arithmetic errors.	Automated day-level proration engine credited via Cashfree PG.
Downgrade Policy	Inconsistent ad-hoc refund credits causing revenue leakage.	Strict No-Refund / Zero-Credit Policy at database layer (₹0.00).
Trainer Telemetry	Siloed logs; coaches unaware when their athletes arrive.	Instant WebSocket alerts pushed to coach upon turnstile entry.
UI & Visualization	Clunky static HTML tables; disjointed user experience.	Interactive 3D Glassmorphism UI (Three.js), Bento HUD.

Feasibility Study

- Technical Feasibility:** High. Built using modern web standards (React 19, Node.js 24, Vite 8, Express 5, MySQL 8). Client-side neural biometric execution eliminates costly dedicated GPU inference servers.
- Economic Feasibility:** Extremely cost-effective. Built entirely on open-source frameworks (MySQL, Socket.io, Face-API), utilizing Cashfree Sandbox API and scalable cloud/on-premise deployment.
- Operational Feasibility:** Seamless user adoption. Provides self-service profile management for coaches/athletes and 1-click administrative review dossiers with complete audit trail persistence.
- Schedule Feasibility:** Delivered on schedule across 14-week phased milestones covering schema design, REST/WebSocket APIs, Cashfree integration, face biometrics, and end-to-end integration testing.

Results & Testing

- Functional Verification:** 100% test pass rate across authentication, biometric verification, membership transitions, and Cashfree gateway webhooks.
- Performance Benchmarks:** Sub-350ms neural facial landmark recognition; sub-20ms MySQL query response time under connection pooling.

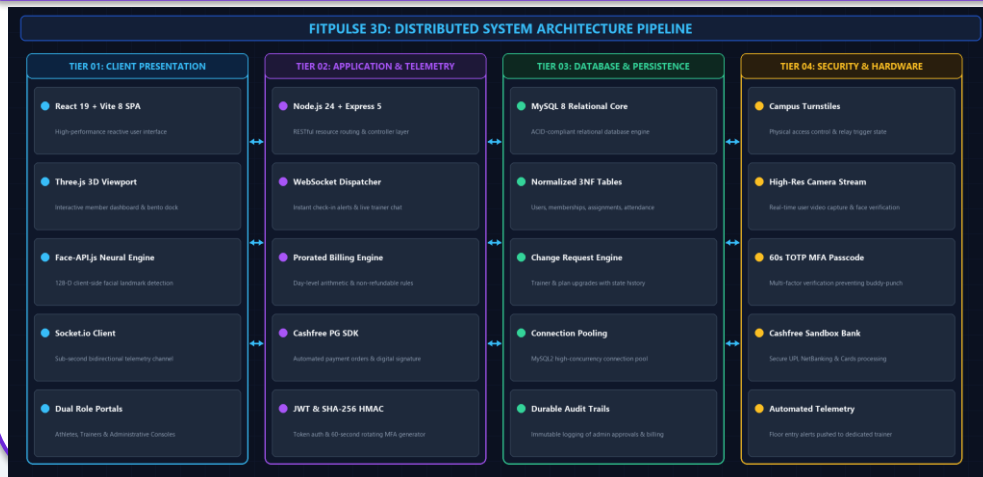
Conclusion

- FitPulse successfully modernizes fitness facility management by bridging physical biometric turnstile security with dynamic financial accounting and automated coaching governance.
- Meets all course outcomes for distributed backend development, secure relational schema design, real-time event streaming, and enterprise software engineering.

Tools & Technologies

- Front-End Stack:** React 19, Vite 8, Three.js (3D Viewport), CSS Glassmorphic Architecture, Socket.io-Client, Vladmandic Face-API.
- Back-End Services:** Node.js 24 (ESM), Express 5, RESTful Endpoints, Cashfree Payment Gateway SDK, Socket.io WebSocket Engine.
- Database Architecture:** MySQL 8 Relational Core (Normalized 3NF Schema, Foreign Keys, Indexing, Connection Pool, ACID Transactions).
- Security & Biometrics:** 128-D Neural Facial Embeddings, SHA-256 HMAC (60s MFA Engine), bcryptjs password hashing, Git, Postman, VS Code, Sharp.

System Architecture



ER Diagram & Database Design



Implementation Screenshots

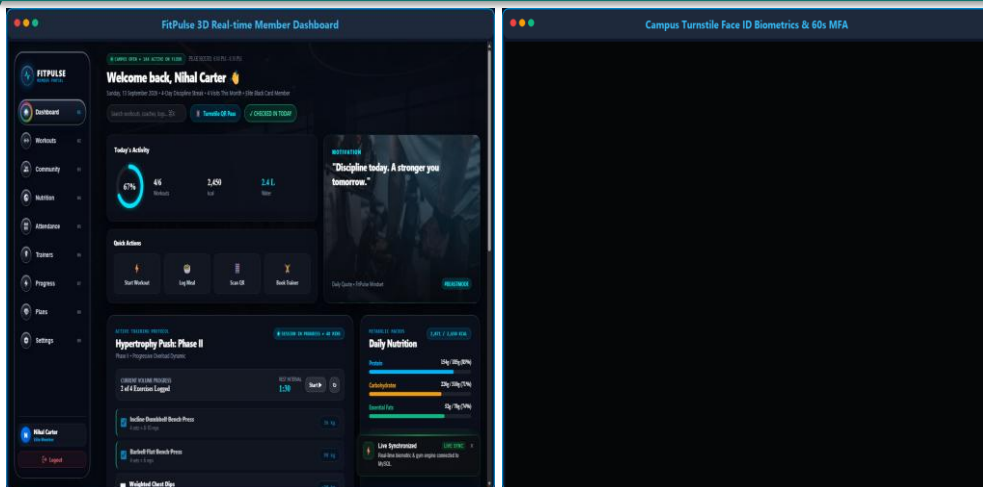


Fig 1: Member 3D Bento Dashboard

Fig 2: Turnstile Face ID & 60s MFA Scanner

Future Scope

- AI Form Correction:** Real-time camera pose estimation (MediaPipe/MoveNet) for barbell lift mechanics and injury prevention.
- IoT Turnstile Relays:** Direct microcontroller interfacing (ESP32 / Raspberry Pi relay gates) for physical turnstile solenoids.

